

5. The ammonium ion contains a 'coordinate bond'. Explain what is meant by this term. [1]

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6. Uranium is used in nuclear fuel reactors. One of its isotopes, uranium-235, has a half-life of 7.03×10^8 years and decays by α -emission.

- (a) Give the mass number and symbol of the element formed as a product of the radioactive decay of uranium-235. [1]

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- (b) If a quantity of uranium-235 decays, state what fraction is left after 2.812×10^9 years. [1]

fraction left =

7. Calculate the mass of calcium that contains the same number of atoms as there are molecules in 9.1 g of sulfur dioxide, SO_2 . [2]

calcium mass = g

